

Problem Set – Loops & Functions

1. Enter destination city, miles travelled to get there and gallons of gasoline used for any number of trips entered at the keyboard (use ctrl+z to stop). Use a function to compute miles per gallon. Display the destination city and miles per gallon for each trip entered. Sum the miles travelled and give a count of the number of trips made. Display these at the end of the program.

Input	Process	Output
• Destination City • Miles Travelled • Gallons Used	<i>computeMpg(Miles Travelled, Gallon Used)</i> • Declare Mpg • Calculate Mpg by finding the quotient of Miles Travelled and Gallons Used • Return Mpg	<i>Each Entry</i> • Destination City • Mpg
	<i>Main()</i> • Declare Mpg, Destination City, Miles Travelled, and Gallons Used • Declare Total Miles and Number of Trips, set at 0 • Assign a variable to the input for Destination City, Miles Travelled, and Gallons Used • While not EOF • <u>Call <i>computeMpg</i></u> to set as Mpg • Add Miles Travelled to Total Miles • Add 1 to Number of Trips • Display Destination City and Trip-Mpg • Ask for new entry for Destination City, Miles Travelled, and Gallons Used • End Loop • Display Total Miles and Number of Trips	<i>After all Entries</i> • Total Miles • Travelled • Number of Trips

2. Allow the employee to enter last name, job code and hours worked (use ctrl+z to stop). Use a function to calculate pay. (Job code L is \$25/hr, A is \$30/hr and J is \$50/hr). Give time and a half for overtime. Display last name and pay for each employee. Sum the pay for each employee as well as count the entries made. After all entries are made, compute and display the average pay and the number of entries made.

Input	Process	Output
- Employee's Last Name - Job Code - Hours Worked	<i>computePay(Job Code, Hours)</i> - Declare Pay, Pay Rate and Overtime - If Job Code equals L, set Pay Rate at \$25 - If Job Code equals A, set Pay Rate at \$30 - If Job Code equals J, set Pay Rate at \$50 - Otherwise, End Program - If Hours Worked is greater than 40, calculate Overtime by finding the difference of Hours and 40, then calculate Pay by finding the sum of 40 * Pay Rate and the product of Overtime and Pay Rate * 1.5 - Otherwise, calculate Pay by finding the product of Hours and Pay Rate - Return Pay	<i>Each Entry</i> - Employee's Last Name - Job Code - Hours Worked - Pay
	<i>Main()</i> - Declare Pay, Employee's Last Name, Job Code, and Hours Worked - Declare Average Pay and Number of Employees, set at 0 - Assign a variable to the input for Employee's Last Name, Job Code, and Hours Worked While not EOF <u>Call <i>computePay</i></u> and set as Pay - Add Pay to Average Pay - Add 1 to Number of Employees - Display Employee's Last Name, Job Code, Hours Worked, Pay - Ask for new entry for Employee's Last Name, Job Code, and Hours Worked End Loop - Calculate Average Pay by finding the quotient of Average Pay and Number of Employees	<i>After all Entries</i> - Average Pay - Number of Employees

	• Display Average Pay and Number of Employees	
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3. Allow any number of students to enter their last name and the credits taken (use `ctrl+z` to stop). Charge \$250 per credit hour. Use a function to compute total tuition. Display student last name, credits taken and tuition owed. Sum tuition and give a count of the number of students who entered data.

Input	Process	Output
• Student's Last Name • Credit Hours • District Code	<i>computeTuition(Credit Hours, District Code)</i> • Declare Tuition, Credit Hour Rate • If District Code equals I (In-District), set Credit Hour Rate at \$250 ○ If District Code equals O (Out of District), set Credit Hour Rate at \$550 ○ Otherwise, End Program • Calculate Tuition by finding the product of Credit Hours and Credit Hours Rate • Return Tuition	<i>Each Entry</i> • Student's Last Name • Tuition
	<i>Main()</i> • Declare Student's Last Name, Credit Hours, District Code, and Tuition • Declare Total Tuition, Total Credit Hours, and Number of Students, set at 0 • Assign a variable to the input for Student's Last Name, Credit Hours, and District Code • While not EOF • <u>Call <i>computeTuition</i></u> and set as Tuition • Add Tuition to Total Tuition • Add Credit Hours to Total Credit Hours • Add 1 to Number of Students • Display Student's Last Name and Tuition • Ask for new entry for Student's Last Name, Credit Hours, and District Code • End Loop • Display Total Tuition, Total Credit Hours, and Number of Students	<i>After All Entries</i> • Total Tuition • Total Credit Hours • Number of Students

4. Any number of customers will enter a product code (W, C, G) and a quantity (ctrl+z to stop). Use a function to determine unit price. Write another function to compute shipping. Then compute the total. Display the product code, unit price, shipping, extended price (quantity x unit price) and total for the order for each entry. Sum and display the total of all entries made.

Product Code	Unit Price	Shipping
W	\$10.00	\$2.00
C	\$15.00	\$5.00
G	\$20.00	\$7.00

Input	Process	Output
- Product Code (W,C,G) - Quantity	<i>computeUnitPrice(Product Code)</i> - Declare Unit Price - If Product Code equals W, set Unit Price at \$10.00 - If Product Code equals C, set Unit Price at \$15.00 - If Product Code equals G, set Unit Price at \$20.00 - Otherwise, End Program - Return Unit Price	<i>Each Entry</i> - Product Code - Unit Price - Shipping - Extended Price - Total
	<i>computeShipping(Product Code)</i> - Declare Shipping - If Product Code equals W, set Shipping at \$2.00 - If Product Code equals C, set Shipping at \$5.00 - If Product Code equals G, set Shipping at \$7.00 - Otherwise, End Program - Return Shipping	<i>After all Entries</i> All Entries Total
	<i>Main()</i> - Declare Product Code, Quantity, Unit Price, Shipping, Extended Price, and Total - Declare All Entries Total, set at 0 - Assign a variable to the input for Product Code and Quantity While not EOF - Call <i>computeUnitPrice</i> and set as Unit Price - Call <i>computeShipping</i> and set as Shipping	

	• Calculate Extended Price by finding the product of Quantity and Unit Price • Calculate Total by finding the sum of Extended Price and Shipping • Add Total to All Entries Total • Display Product Code, Unit Price, Shipping, Extended Price, and Total • Ask for new entry for Product Code and Quantity • End Loop • Display All Entries Total	
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5. Allow students to enter the department and course code as noted below for any number of courses (ctrl+z to stop). Use a function to determine the lab fee also in the table below. For each entry display the department, course code and lab fee. Give the total of all lab fees to collect. Compute and display the average lab fee.

Department	Course Code	Lab Fee
CIS	101	\$50.00
CIS	121	\$100.00
MAT	111	\$25.00
MAT	112	\$35.00
ENG	100	\$55.00
All Others		\$50.00

Input	Process	Output
• Department • Course Code	<i>computeLabFee(Department, Course Code)</i> • Declare Lab Fee • If Department equals CIS and Course Code equals 101, set Lab Fee at \$50.00 ○ If Department equals CIS and Course Code equals 121, set Lab Fee at \$100.00 ○ If Department equals MAT and Course Code equals 111, set Lab Fee at \$25.00 ○ If Department equals MAT and Course Code equals 112, set Lab Fee at \$35.00	<i>Each Entry</i> • Department • Course Code • Lab Fee

	○ If Department equals ENG and Course Code equals 100, set Lab Fee at \$55.00 ○ If Department either equals CIS, MAT, or ENG, set Lab Fee at \$50.00 ○ Otherwise, End Program ● Return Lab Fee	
	<i>Main()</i> ● Declare Department, Course Code, and Lab Fee ● Declare Total Lab Fees, Number of Courses, and Average Lab Fee, set at 0 ● Assign a variable to the input for Department and Course Code ● While not EOF ● <u>Call <i>computeLabFee</i></u> and set as Lab Fee ● Add Lab Fee to Total Lab Fees and Average Lab Fee ● Add 1 to Number of Courses ● Display Department, Course Code, Lab Fee ● Ask for new entry for Department and Course Code ● End Loop ● Calculate Average Lab Fee by finding the quotient of Average Lab Fee and Number of Courses ● Display Total Lab Fees and Average Lab Fee	<i>After all Entries</i> ● Total Lab Fees ● Average Lab Fee